

IDENTITY IS THE SIGNAL: WHY PRE-EXECUTION AGENT-ACTION RISK NEEDS DELEGATED AUTHORITY, NOT JUST CONTENT

Anonymous authors

Paper under double-blind review

ABSTRACT

Guardrail models for AI agents typically classify a proposed action from its text alone. We show this misses the dimensions that matter most for delegated, tool-using agents. Stripping principal entitlements, delegated scopes, and policy text from a trained 2B-parameter decoder’s input drops macro AUPRC from 0.915 to 0.736 on held-out tools, concentrated entirely in three dimensions — unauthorized-scope detection falls from 1.00 to 0.12, policy-conflict detection from 1.00 to 0.35 — while two zero-shot safety models (Granite Guardian 3.3 8B, gpt-oss-safeguard 20B) sit at chance on exactly these dimensions with or without that context, showing the gap is not something prompting a general safety model can close. We further show that the *shape* of policy-conditioned reasoning, not just tool identity, must generalize to unseen policy kinds and phrasings, and that naive multi-policy composition collapses to a 0.97 false positive rate by three stacked policies unless decisions are made from calibrated joint probabilities rather than independent thresholds. An independent blind LLM judge, shown only the rendered text, confirms two of the three dimensions the identity-ablation result rests on (policy-conflict, unauthorized-scope) and finds the third (insufficient-context) unrecoverable from the text at all, which we report as an open validity gap. On AgentDojo banking traces we did not generate, zero-shot, the 8B checker reaches 0.80 AUPRC on prompt-injection influence against a 0.15 base rate and the 2B checker reaches 0.85 AUPRC on policy conflict against a 0.24 base rate, both against rule baselines at or near base rate, while the 2B checker fails outright on injection. All results are on synthetic data; we are explicit about what that does and does not license.

1 INTRODUCTION

Deployed AI agents increasingly act with delegated authority: a support agent issues refunds for a merchant, a coding agent pushes commits for a developer, an operations agent grants IAM roles for an administrator. Before such an action executes, some system must decide whether it is safe to proceed. The dominant tool is a guardrail classifier that reads the proposed action and outputs a harm judgment (Inan et al., 2023; Padhi et al., 2024; Zeng et al., 2024; Ghosh et al., 2024; OpenAI, 2025).

We argue this conflates two questions. **Content risk** asks whether an action looks dangerous on its face (deletes data, moves money, contacts an external party) and is answerable from the action text alone. **Authorization risk** asks whether *this* action is something *this* agent was allowed to do, on behalf of *this* principal, under the policy in force – and requires the principal’s entitlements, the agent’s delegated scope, and the applicable policy clause as structured inputs, not prose the model must infer.

We build a pre-execution risk classifier, *forecheck*, whose input contract makes these fields explicit, then strip them from the same trained model’s input to see what breaks. On synthetic data, three of eleven risk dimensions collapse from near-perfect to near-chance when identity is removed, while dimensions already recoverable from action text alone are unaffected. This is not a fine-tuning artifact prompting could fix: two zero-shot safety models given the same policy text in-prompt are

at chance on exactly these dimensions, with or without identity, because they were never trained to use it. We further test whether policy-conditioned reasoning generalizes to unseen policy *kinds* and *phrasings*, and show that composing several such classifiers’ decisions – how guardrail stacks are deployed in practice – fails in a specific, fixable way: per-policy thresholding compounds toward denying everything as more policies are added, while deciding from the joint calibrated probability vector does not. On AgentDojo banking traces we did not generate, zero-shot, the 8B checker reaches 0.80 AUPRC on prompt-injection influence against a 0.15 base rate and the 2B checker reaches 0.85 AUPRC on policy conflict against a 0.24 base rate, both against rule baselines at or near base rate, while the 2B checker fails outright on injection.

Contributions. (1) An identity-ablation diagnostic separating which of eleven risk dimensions need delegated-authority context from which are content lookups (§3); (2) evidence that two current zero-shot safety models do not use delegated-authority context even when it is handed to them (§3); (3) a policy-generalization result showing robustness to unseen policy wording but not unseen policy kinds (§4); (4) a multi-policy composition result showing calibrated joint-probability decisions, not rule engineering, keep a guardrail stack from collapsing to "deny everything" (§5); (5) a comparison showing an agent asked to judge its own action cannot recover the authorization signal an external checker can (§6); and (6) an independent blind-judge check of whether the generator’s labels are recoverable from the rendered text at all, which confirms the identity-ablation result on two of its three load-bearing dimensions and flags the third as unvalidated (§7); (7) a zero-shot transfer result on AgentDojo, a published external benchmark, showing the policy-conflict and prompt-injection results transfer to agent traces and policies forecheck never saw, while unauthorized-scope does not and calibration does not (§8). Every number here except §8 is on synthetic data from forecheck’s own scenario generator; §10 states plainly what that does and does not license.

2 PROBLEM SETTING

Contract. forecheck scores a proposed tool call against an `ActionContext`: the principal’s entitlements, the scope actually delegated to the agent (a possibly strict subset of the principal’s), a trust-labelled trajectory of prior observations, the applicable policy clause(s), and the proposed action (tool, arguments, destination) – typed fields the classifier reads directly, not facts inferred from prose. A classifier that sees only the rendered action text, as every guardrail model in §9 does, can answer content-risk questions but has no way to check whether the agent was delegated the scope it is exercising.

Eleven risk dimensions. Four are *content/consequence* dimensions decidable from the action and destination alone: `destructive_or_irreversible_action`, `financial_commitment`, `external_communication`, `untrusted_destination`. Two are *trajectory* dimensions decidable from prior observations without identity: `prompt_injection_influence`, `suspicious_action_sequence`. The remaining are where identity and policy are load-bearing: `unauthorized_scope` (inside the principal’s entitlements *and* the agent’s delegated scope?), `policy_conflict` (violates a stated clause?), `insufficient_context` (enough delegated information to decide at all?), and `privilege_escalation` (does the action widen authority beyond what was granted?). `sensitive_data_exposure` behaves like a content dimension in practice (§3) because its deciding field renders directly into the action arguments.

Data. All results are on synthetic data from forecheck’s own scenario generator (49,996 rows at the v2/v3/v4 scale used here), split into `train/calibration/ dev/test`, a `heldout_family` split withholding a stratified subset of tools, and, from v3 on, `heldout_policy_kind/heldout_policy_phrasing` splits (§4). Every number measures whether an arm learned the generator’s label function, not real-world risk; no human-labelled or production-traffic evaluation exists at submission time (§10).

Arms. A LoRA-adapted 2B decoder (`granite-3.3-2b-instruct`) scored by candidate-token logits over the eleven dimensions (Hu et al., 2021); two linear-head encoders (ModernBERT-large (Warner et al., 2024), `granite-embedding-r2`); a deterministic rule baseline with no learned component; two zero-shot safety models with the same policy text in-prompt and no fine-tuning, Granite Guardian 3.3 8B (Padhi et al., 2024) and `gpt-oss-safeguard 20B` (OpenAI, 2025); and an *agent self-*

Table 1: Macro AUPRC with and without identity/policy context. Decoder and encoder rows are on heldout_family unless noted; zero-shot rows use a 2,000-row (Granite Guardian) or 400-row (gpt-oss-safeguard) subsample. Zero-shot models are unaffected or slightly helped by stripping, showing they never used the context. Source: docs/results/synthetic-v2/README.md.

Arm	Split	Macro AUPRC (full)	Macro AUPRC (stripped)
Decoder 2B v2	heldout_family	0.915	0.736
Decoder 2B v4	heldout_family	0.960	0.790
Decoder 8B v4	heldout_family	0.958	0.786
Decoder 2B v4, trained stripped	heldout_family	—	0.784
Encoder ModernBERT-large	heldout_family	0.813	0.712
Encoder granite-embedding-r2	heldout_family	0.841	—
Granite Guardian 3.3 8B (zero-shot)	heldout_family	0.257	0.271
gpt-oss-safeguard 20B (zero-shot)	test (n=400)	0.269	0.238
Agent self-judgment 8B ^a	heldout_family (n=2000)	0.242	0.281
Rule baseline	heldout_family	0.614	0.614

^aSingle ALLOW/STOP verdict scored against all eleven labels (§6), not a per-dimension classifier.

judgment condition where an untrained instruct model (2B, 8B (IBM Granite Team, 2024)) gets the same context in a Fin-style system prompt and returns one ALLOW/STOP verdict (§6).

3 IDENTITY ABLATION

At evaluation time only, we remove principal entitlements, delegated scope, on_behalf_of, and policy text from the trained decoder’s rendered context (`-strip-identity`), leaving action, destination, and trajectory untouched, and compare against a rule baseline (action and destination fields only) and two zero-shot safety models given the same policy text in-prompt.

Table 1: identity-aware trained models lose 0.10–0.18 macro AUPRC when identity is removed; the rule baseline is unchanged; the zero-shot models are flat or slightly *improve*. If a general safety model quietly used delegated-authority context, stripping it would hurt – it does not, for either model.

The v4 ablation replicates the v2 pattern on the fixed data: macro AUPRC drops 0.960 to 0.790, close to v2’s 0.915 to 0.736. The one change is `privilege_escalation`, which now survives stripping entirely (1.000 to 1.000): the v4 render fix makes `authority_before/authority_after` observable directly in the tool-call text, so it is a content dimension, not an identity one. An 8B decoder on the same v4 data shows the identical collapse (0.958 to 0.786, Table 1), with the same three dimensions falling and the same two surviving – model scale gives the classifier no way to infer entitlements or policy text that were simply not in its input.

Training on data rendered without identity fields from the start, rather than stripping only at evaluation time, gives the same ceiling: macro AUPRC 0.780/0.784 on test/heldout_family, within a point of the eval-time-stripped model’s 0.787/0.790. The three delegated-authority dimensions still do not recover: `heldout_family unauthorized_scope` falls to 0.107, `policy_conflict` to 0.429, `insufficient_context` to 0.357, matching or below the eval-time-stripped model and nowhere near the full model’s 0.999/0.964/0.841. This rules out the reading that eval-time stripping merely broke a fragile mechanism the model had learned to use – the information is absent from the rendered content itself.

Figure 1 shows the same pattern across all eleven dimensions: the drop from stripping identity is concentrated in `unauthorized_scope`, `policy_conflict`, and `insufficient_context`, while the remaining dimensions barely move.

The macro drop is concentrated entirely in the three delegated-authority dimensions. `unauthorized_scope` falls below the rule baseline (0.116 vs. 0.512) once the entitlements it must be checked against are gone; `policy_conflict` falls to 0.345, barely above the 0.315 rule floor, because there is no clause left to violate; `insufficient_context` falls by half. The fourth row is untouched: its deciding fact renders directly into the action

Table 2: Decoder 2B v2, heldout_family AUPRC per dimension: full, identity-stripped, and rule baseline. First three rows are delegated-authority dimensions; the last is a representative content/lookup dimension. Source: docs/results/synthetic-v2/README.md.

Dimension	Decoder (full)	Decoder (stripped)	Rule baseline
unauthorized_scope	1.000	0.116	0.512
policy_conflict	0.999	0.345	0.315
insufficient_context	0.849	0.418	0.235
financial_commitment (lookup)	1.000	1.000	1.000

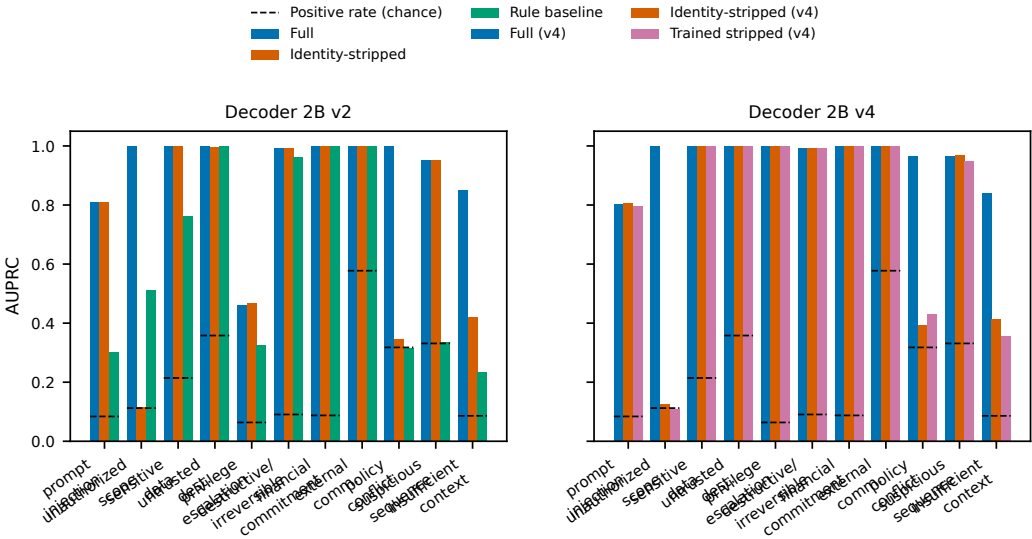


Figure 1: Decoder 2B v2, per-dimension AUPRC on heldout_family ($n=6640$, synthetic data): full model vs. identity-stripped vs. rule baseline, with each dimension’s positive rate (chance level) marked by a dashed line. Source: docs/results/synthetic-v2/decoder-2b/.

arguments regardless of identity, and the rule baseline already gets it right. Four more dimensions (`destructive_or_irreversible_action`, `untrusted_destination`, `external_communication`, `sensitive_data_exposure`) show the same flat, lookup pattern; two more (`prompt_injection_influence` 0.810, `suspicious_action_sequence` 0.952) are learned from trajectory order and are identity-independent.

The zero-shot models’ near-chance macro AUPRC (0.24–0.27), unmoved by identity, shows this is not a prompting gap: both were trained for harm, jailbreak, and groundedness criteria, not for checking an action against entitlements or policy. `gpt-oss-safeguard` is also $\sim 50\times$ slower per row than the 2B decoder, for no benefit on the dimensions that matter here.

This whole claim rests on the three delegated-authority dimensions being real labels, not generator artifacts an independent reader cannot recover; §7 checks that directly and finds the claim solid for two of the three, with `insufficient_context` unvalidated.

4 POLICY GENERALIZATION: KIND VS. PHRASING

Robustness to unseen *tools* (`heldout_family`) is not robustness to unseen *policy kinds* or *phrasings*, and the latter is the more product-relevant test: a deployment needs the classifier to apply a customer’s own wording, not just handle a new tool under a training-set policy. ADR 0010 adds two splits: `heldout_policy_kind` withholds entire predicate kinds (e.g. `forbid_recipient_domain`) from training; `heldout_policy_phrasing` withholds one paraphrase index of every kind the model *did* train on. v4 additionally fixes a renderer defect that left `privilege_escalation` undecidable from text (§10).

Figure 2 makes the kind-vs-phrasing asymmetry visible: bars for test, `heldout_family`, and `heldout_policy_phrasing` sit close together near the ceiling, while `heldout_policy_kind` drops toward, though still above, the rule baseline line, for every generator and both model sizes.

Unseen phrasing generalizes. A paraphrase index never seen in training leaves `policy_conflict` at 0.980–0.993 AUPRC, matching the in-distribution test split (0.990–0.996): the model reads the clause’s meaning, not a template id. **Unseen policy kinds do not, even with twice as many kinds.** ADR 0011’s v5 triples the kind inventory (15 to 30) while withholding the same proportion (4 of 30 vs. 2 of 15). Across three training seeds, `policy_conflict` on `heldout_policy_kind` lands at 0.704, 0.755, and 0.717 AUPRC (mean 0.725, sample sd 0.026, $n=3$) – essentially unchanged from the v4 15-kind recipe’s own three-seed band of 0.752, 0.689, and 0.711 (mean 0.717, sample sd 0.032, $n=3$; §10). The v5 mean sits within one sample sd of the v4 mean, so tripling the kind inventory is not a real improvement, it is inside the seed noise the 15-kind recipe already showed. The rule baseline on this split is 0.325, so the model is still far above chance; it has just not learned the general operation. **Model size helps, partially.** An 8B LoRA on the unchanged v4 data reaches 0.851 AUPRC (AUROC 0.916) on the same `heldout_policy_kind` `policy_conflict` cell – a real gain over both 2B bands – but still well short of the ~ 0.98 phrasing-generalization ceiling both sizes reach. Capacity recovers some of the missing generalization; it does not close the gap.

Table 3: Decoder 2B, macro AUPRC and `policy_conflict` (pc) AUPRC across splits, on the v3 (ADR 0010, 15 policy kinds, 2 withheld), v4 (same data, `privilege_escalation` render fix), and v5 (ADR 0011, 30 kinds, 4 withheld) generators. Last column: an 8B LoRA on the unchanged v4 data, `policy_conflict` only. *v5 `heldout_policy_kind` pc is seed 0 of three training seeds (mean 0.725, discussed in text). Source: docs/results/synthetic-v2/README.md.

Split	v3 macro	v4 macro	v5 macro	v3 pc	v4 pc	v5 pc	8B v4 pc
test	0.908	0.960	0.957	0.994	0.994	0.996	0.990
<code>heldout_family</code>	0.914	0.960	0.951	0.941	0.964	0.933	0.965
<code>heldout_policy_kind</code>	0.889	0.944	0.941	0.653	0.752	0.704*	0.851
<code>heldout_policy_phrasing</code>	0.921	0.968	0.970	0.987	0.993	0.980	0.978

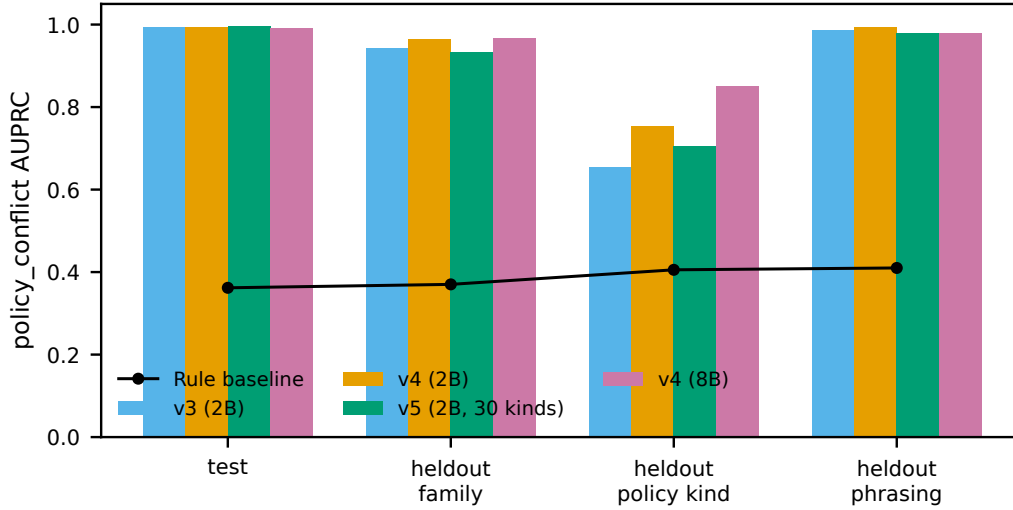


Figure 2: Decoder 2B, `policy_conflict` AUPRC across splits for the v3, v4, and v5 generators, plus the 8B LoRA on v4 data (synthetic data; n per split matches Table 3), with the rule baseline’s `policy_conflict` AUPRC on each split shown as a line. Source: docs/results/synthetic-v2/decoder-2b-v3/, decoder-2b-v4/, decoder-2b-v5/, decoder-8b-v4/.

The v4 fix also confirms `privilege_escalation`’s earlier AUROC-high/AUPRC-low signature (§10) was a data defect, not a model limit: rendering the deciding `authority_before/authority_after` values lifts it from 0.505 to 1.000 AUPRC on `heldout_family`.

5 MULTI-POLICY COMPOSITION AND THE CALIBRATION HINGE

Guardrail deployments commonly run several independently-tuned policies at once and act on the most restrictive decision – OR-ing k binary guards. Each guard’s false-positive rate compounds under that OR purely as a function of k , independent of any guard getting worse. `forecheck`’s alternative is one calibrated probability vector over the eleven dimensions, composed by one policy engine, decided either by per-dimension thresholds (`threshold` mode) or by the argmin of expected cost over the joint vector (`expected_cost` mode).

Figure 3 shows the independent and joint threshold strategies’ FPR climbing sharply as k grows past 2, while `expected_cost_joint` stays flat; the encoder’s `expected_cost_joint` curve, by contrast, sits near the top of the FPR panel throughout, illustrating that the flat curve is a property of the decoder’s calibration, not of the decision rule alone.

Table 4: Decoder 2B, false positive rate on covered-benign examples, `heldout_family` ($n=6640$), stacking k synthetic per-dimension policies. `independent` ORs k separately-thresholded decisions; `expected_cost_joint` merges the same k policies into one bundle and decides from the joint probability vector. Full table with the `joint` strategy, FNR, review rate, and the granite-embedding-r2 encoder is Table 9 in the appendix. Source: docs/evaluation/multi-policy-stacking.md.

k	1	2	3 (8 identical)
independent FPR	0.015	0.140	0.966
expected_cost_joint FPR	0.015	0.082	0.082

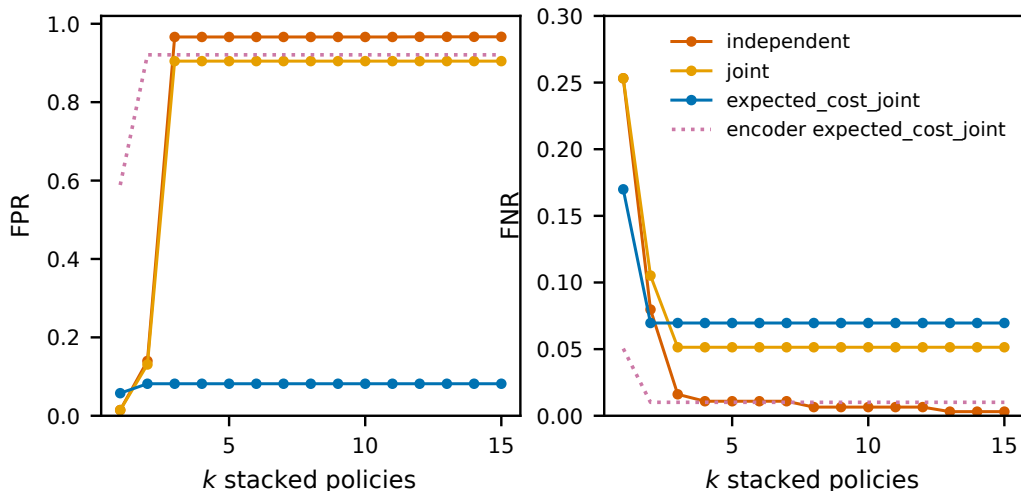


Figure 3: Decoder 2B, FPR (left) and FNR (right) vs. number of stacked policies k on `heldout_family` ($n=6640$, synthetic data), for independent, joint, and `expected_cost_joint` strategies; the granite-embedding-r2 encoder’s `expected_cost_joint` curve is shown dotted for comparison. Source: `docs/results/synthetic-v2/decoder-2b/heldout_family-stacking-report.md`, `encoder-granite-embedding-r2/heldout_family-stacking-report.md`.

Threshold-based independent stacking pushes the decoder’s FPR from 0.015 at $k=1$ to 0.966 by $k=3$, staying there through $k=8$ – effectively denying everything, not because any policy got worse but purely because more were added. Deciding from the joint calibrated probability vector under `expected_cost` is invariant to k from $k=2$ on (FPR/FNR/review rate flat at 0.082/0.070/0.28): the decision depends on the probability vector and a fixed cost table, not on how many rules mention a dimension, so stacking more policies cannot by itself make the classifier more conservative.

The invariance is a calibration property, not a decision-rule property. The same rule on the granite-embedding-r2 encoder’s probabilities – identical examples and policies – settles at FPR 0.921, not 0.08 (Table 9). The decoder is well calibrated (macro ECE 0.004 on `heldout_family`, temperature-scaled on the calibration split (Guo et al., 2017; Platt, 1999)); the encoder is flatter (ECE 0.049). Expected-cost composition only keeps a stack from collapsing when the probabilities feeding it are sharp and calibrated, and cannot be recovered by rule engineering alone on a poorly-calibrated model.

The cost table used throughout is the default severity-tier table, not tuned to any deployment: the $k=1$ row already trades false positives for false negatives relative to the F1-optimal threshold (FPR/FNR 0.058/0.170 vs. 0.015/0.253). The k -invariance claim is about the *shape* of the composition curve, not that this default cost table is right for any given customer.

6 CAN THE ACTING AGENT JUDGE ITS OWN SCOPE?

The classifier above is an external checker: it never generates the action, only scores it. A cheaper alternative is to ask the acting agent itself, from inside its own system prompt, whether to proceed. We give the same untrained instruct models used as the decoder’s base (2B, 8B) a Fin-style system prompt carrying the identical entitlements, delegated scope, and policy text, followed by the pending tool call, and read a single ALLOW/STOP verdict off the first output token. That one scalar is scored against all eleven dimensions’ labels: tracking the external checker means an undifferentiated act/refuse decision happens to catch that risk; lagging badly means the dimension needs external, per-dimension structure the agent’s own judgment does not supply.

Table 5: Agent self-judgment (8B, untrained, single ALLOW/STOP verdict) vs. the trained decoder external checker (v4), AUPRC on heldout_family ($n=2000$ for the agent rows). Positive rate for each dimension in parentheses. Source: docs/results/synthetic-v2/README.md.

Dimension (pos. rate)	Agent 8B	Agent 8B, stripped	Decoder 2B v4 (checker)
unauthorized_scope (0.106)	0.138	0.104	0.999
policy_conflict (0.385)	0.509	0.415	0.964
insufficient_context (0.079)	0.085	0.084	0.841
privilege_escalation (0.075)	0.103	0.106	1.000
prompt_injection_influence (0.075)	0.169	0.498	0.804
Macro	0.242	0.281	0.960

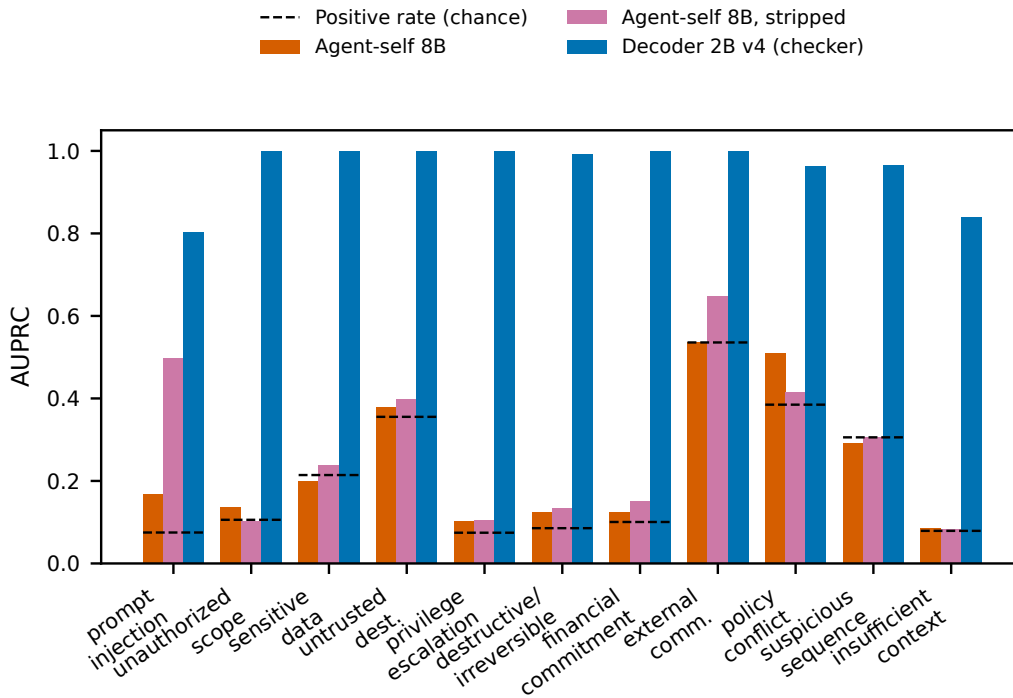


Figure 4: Per-dimension AUPRC on heldout_family (synthetic data; $n=2000$ for the agent-self-8B rows, $n=6653$ for the decoder), with each dimension’s positive rate (chance level) marked by a dashed line. Source: docs/results/synthetic-v2/agent-self-8b/, decoder-2b-v4/heldout_family-report.md.

Figure 4 shows the pattern holds across all eleven dimensions, not just the five in Table 5: the agent’s bars track the chance line closely everywhere except `policy_conflict` and `external_communication`, while the external checker clears 0.84 AUPRC or better on every dimension.

Two findings. First, **the agent cannot judge its own scope from inside its prompt**: with full identity context the 8B agent is at or near chance on every identity-dependent dimension (AUROC 0.53–0.67, not shown), the same level as the zero-shot safety models in §3. The delegation information sits in its own prompt and is not turned into a decision. Second, **identity context makes the agent worse, not better, at spotting injection**: stripping the policy and scope text raises `prompt_injection_influence` AUPRC from 0.169 to 0.498 – a long policy preamble appears to dilute attention to an injected instruction elsewhere in the trajectory. Both argue for keeping an external checker in the loop rather than trusting the agent’s own judgment, even on content-style dimensions. The small identity signal that does exist (`unauthorized_scope` AUROC 0.61 full vs. 0.48 stripped) shows the agent reads the context a little; it does not turn that into a usable decision, and the gap to the external checker’s 0.960 macro AUPRC persists regardless.

Row-for-row comparison on v4 (`agent-self-8b-v4/`). Scoring the decoder 2B v4 checker against the identical 2,000 `heldout_family` rows used for the 8B agent above, it beats the agent’s own single-verdict score on every one of the eleven dimensions, by 0.45–0.90 AUPRC: `unauthorized_scope` 0.999 vs. 0.135, `policy_conflict` 0.964 vs. 0.509, `insufficient_context` 0.841 vs. 0.086, `prompt_injection_influence` 0.804 vs. 0.166 (agent stripped: 0.507). Identity context in the agent’s own prompt still costs it on injection detection even though stripping barely moves the checker; the agent’s own macro is 0.242 full vs. 0.281 stripped.

7 ARE THE LABELS REAL? AN INDEPENDENT BLIND JUDGE

Every number so far measures whether an arm learned forecheck’s own generator’s label function (§10). That leaves open a narrower but answerable question: is the label itself observable in the rendered text at all, to a reader who never sees the generator’s latent scenario? We had a frontier model (`gpt-5.6-sol`), given only the rendered context and the eleven dimension definitions, judge 600 v4 rows stratified so every dimension has at least 20 positive and 20 negative examples, across `test`, `heldout_family`, and `heldout_policy_kind`. Agreement with the generator’s label is not a ground-truth check – the judge is another reader of the same text a classifier sees – but low agreement is informative: it means the label depends on information the generator tracked internally and never rendered, so a trained model scoring well on it is learning a generator regularity, not a fact about the text.

Table 6 and Figure 5 split into three tiers. **Recoverable** ($\kappa > 0.7$): `privilege_escalation`, `prompt_injection_influence`, and `policy_conflict`. A blind reader recovers these from the text alone. `privilege_escalation` at 0.91 is an independent confirmation of the v4 render fix (§4): the deciding fields are now genuinely in the tool-call arguments, not just in the model’s training signal. `policy_conflict` at 0.71 with judge recall 0.89 means the policy clause plus the proposed call is enough for an outside reader to spot the conflict; the disagreements are mostly the judge reading a clause more strictly than the generator’s own predicate, not a missing signal.

Definitional disagreement ($0.2 \leq \kappa \leq 0.5$): `unauthorized_scope`, `financial_commitment`, `destructive_or_irreversible_action`, `suspicious_action_sequence`, `external_communication`, and `untrusted_destination`. Here the judge and the generator are both reading the text and both producing a coherent decision – they just use different definitions of the same dimension name. For `unauthorized_scope` the judge flags roughly twice as many rows as the generator (159 vs. 81 judge-positive) because it treats a mismatch with the agent’s stated objective (e.g. a bulk retrieval where a single-record lookup was requested) as out of scope, while the generator’s label is entitlement-based only; judge recall of the generator’s positives is still 0.74, so the two notions overlap substantially without being identical. For `financial_commitment` the judge requires funds to actually move, while the generator labels any call that carries a financial amount

Table 6: Independent LLM-judge agreement with the generator label, yes vs. not-yes, $n=600$ v4 rows (gpt-5.6-sol, blind to the generator’s latent scenario). “Stripped” re-judges the same 600 rows with the `-strip-identity` rendering used for the model identity ablation (§3). Agreement/recall/precision are for the full-context pass. Source: docs/results/judge/README.md.

Dimension	κ full	κ stripped	Agreement	Judge recall	Judge precision
privilege_escalation	0.909	0.909	0.988	1.00	0.84
prompt_injection_influence	0.905	0.905	0.983	0.84	1.00
policy_conflict	0.714	0.029	0.862	0.89	0.78
destructive_or_irreversible_action	0.511	0.542	0.897	0.60	0.54
suspicious_action_sequence	0.404	0.428	0.852	0.82	0.33
unauthorized_scope	0.391	0.040	0.800	0.74	0.38
financial_commitment	0.359	0.300	0.935	0.24	0.92
external_communication	0.331	0.361	0.897	0.86	0.23
untrusted_destination	0.213	0.205	0.777	0.41	0.29
insufficient_context	0.059	−0.036	0.843	0.14	0.15
sensitive_data_exposure	0.002	0.061	0.783	0.17	0.08

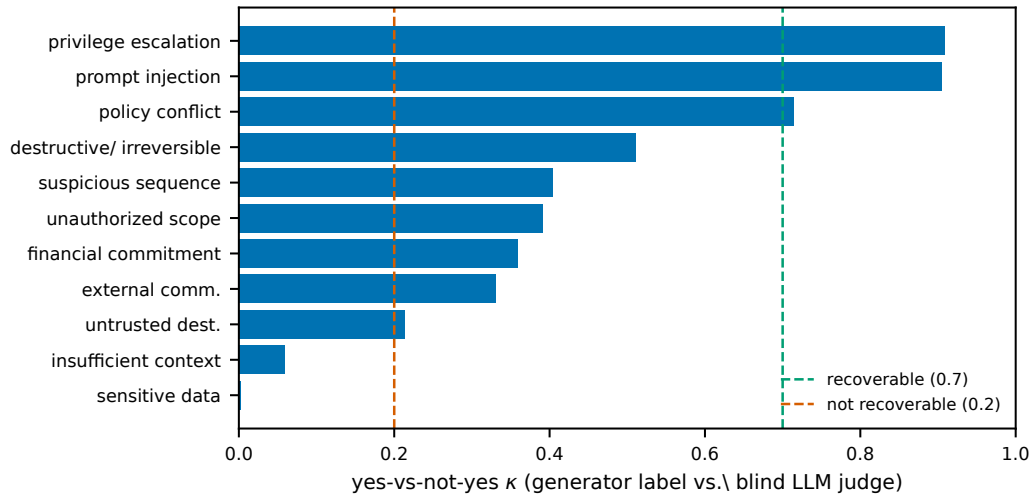


Figure 5: Yes-vs-not-yes κ per dimension, sorted, with the recoverable (0.7) and not-recoverable (0.2) tier lines from §7. Source: docs/results/judge/README.md.

in its arguments. These are taxonomy decisions to fix in the dimension definitions, not evidence the labels are unobservable.

Not recoverable ($\kappa < 0.1$): `insufficient_context` and `sensitive_data_exposure`. For `insufficient_context` the judge calls the context sufficient on 50 of the 58 generator-positive rows in the sample; whatever the generator tracks as a context gap is not visible in what gets rendered. For `sensitive_data_exposure` the judge requires a transfer to a lower-sensitivity destination, while the generator labels any read of confidential data through an already-authorised tool as exposure – recall 0.17, precision 0.08, essentially two different dimensions sharing a name.

Consequence for the identity-ablation claim. §3’s claim that delegated-authority information is load-bearing rests on exactly three dimensions: `unauthorized_scope`, `policy_conflict`, and `insufficient_context`. Judged independently, that claim is solid on `policy_conflict` ($\kappa=0.71$), directionally confirmed but not identical in definition on `unauthorized_scope` ($\kappa=0.39$, judge recall 0.74), and unvalidated on `insufficient_context` ($\kappa=0.06$): a blind frontier reader cannot recover this label from the text at all, which means the trained decoder’s 0.849 AUPRC on it (Table 2) is closer to learning a generator regularity than to answering a checkable question about the rendered context. We report `insufficient_context` numbers throughout this paper because the identity-ablation *pattern* – three dimensions collapse together, four content dimensions do not move – still holds with it removed on the other two legs alone; but any claim resting on `insufficient_context` specifically should be read as unconfirmed until the generator makes its context-gap condition explicit in the rendering. The two content dimensions that survive identity stripping in §3 (`prompt_injection_influence`, `privilege_escalation`) are exactly the two the judge recovers best, consistent with them being genuine properties of the text rather than generator lookups.

Judging without identity context. We re-ran the same judge on the same 600 rows with identity stripped from the rendered context – the `-strip-identity` rendering used in §3 – to test whether a blind reader loses the same dimensions the trained checker loses. It does, and only those two. `policy_conflict`’s κ falls from 0.714 to 0.029: the judge finds 28 positives instead of 259 because the policy text it would check the action against is no longer in the context. `unauthorized_scope`’s κ falls from 0.391 to 0.040: the judge now flags 203 rows at precision 0.16, which means it is guessing from the action alone rather than reading an entitlement. Every other dimension is unchanged within noise (Table 6); the largest other move is `destructive_or_irreversible_action`, 0.511 to 0.542. `insufficient_context` stays unrecoverable in both passes (0.059 and -0.036), the third leg of the identity-ablation claim that remains unvalidated. This is an independent replication of the identity-ablation result on labels rather than on model scores: a blind frontier reader loses the same two dimensions the trained checker loses when identity is stripped (`unauthorized_scope` 0.999 to 0.125 AUPRC, `policy_conflict` 0.964 to 0.392 AUPRC, Table 2), and keeps the same content dimensions (`prompt_injection_influence`, `privilege_escalation`) the checker keeps. The content dimensions’ judge-yes and generator-yes counts are identical between the two passes, which also confirms that stripping identity left the tool-call and trajectory text itself unchanged.

Scoring the checker against the judge’s labels. Re-scoring the trained 2B v4 decoder and the rule baseline against the judge’s labels on the judged rows ($n=257$ on `heldout_family`) gives Table 7. Against labels the generator did not write, the checker keeps the three recoverable dimensions and `untrusted_destination`, and beats the rule baseline on each by 0.08–0.68 AUPRC; macro 0.61 vs. 0.42. `unauthorized_scope` drops to 0.59 for the definitional reason above (the judge counts a mismatch with the stated objective as out of scope; the trained label is entitlement-based). `insufficient_context` at 0.09 confirms the label is a generator artefact. The lookup dimensions collapse for both arms because the judge defines them differently, so they are not evidence for or against either.

Table 7: AUPRC against the independent judge’s labels (gpt-5.6-sol) on the judged heldout_family rows ($n=257$), decoder 2B v4 vs. rule baseline, with the decoder’s AUPRC against the generator’s own labels on the full split for reference. Source: docs/results/judge/README.md.

Dimension	Decoder 2B v4	Rule baseline	Decoder vs. generator labels
prompt_injection_influence	1.000	0.459	0.804
privilege_escalation	0.977	0.297	1.000
policy_conflict	0.889	0.502	0.964
untrusted_destination	0.864	0.785	1.000
destructive_or_irreversible_action	0.635	0.509	0.992
unauthorized_scope	0.592	0.509	0.999
suspicious_action_sequence	0.528	0.467	0.966
external_communication	0.437	0.381	1.000
sensitive_data_exposure	0.405	0.291	1.000
financial_commitment	0.327	0.280	1.000
insufficient_context	0.090	0.090	0.841
macro	0.613	0.415	0.960

8 TRANSFER TO AGENTDOJO: DATA WE DID NOT GENERATE

Every result so far is on forecheck’s own synthetic generator (§10). We score the same v4 checkers, zero-shot, on AgentDojo (Debenedetti et al., 2024), a published benchmark of agent run traces we did not write. We use the banking suite traces for claude-3-5-sonnet-20241022 and gpt-4o-2024-05-13: 1,714 runs, 4,215 proposed tool calls, 11 attack types plus no-attack runs. Labels come from AgentDojo’s own ground_truth() functions, not from forecheck’s generator: prompt_injection_influence is yes when a call executes the injection task’s goal (positive rate 0.150, $n=4,133$ evaluable calls); unauthorized_scope is yes when the call’s function is outside the user task’s reference-solution function set (rate 0.253); policy_conflict is yes when one of four banking policies we wrote once per suite fires, checked by deterministic predicates (rate 0.237); financial_commitment (0.343) and destructive_or_irreversible_action (0.097) come from function-name sets. Trace parsing and the full label derivation are documented in docs/external-agentdojo.md. Every arm is zero-shot: nothing was trained or tuned on AgentDojo, the calibration bundles were fitted on synthetic dev, and threshold selection is disabled.

Table 8 and Figure 6 give six readings. **Policy checking transfers.** 2B v4 scores 0.849 AUPRC on policy_conflict against a 0.237 base rate and a rule baseline at base rate, on policies, a tool schema, and agent traces it never saw. Eval-time stripping drops it to 0.694 and the train-time stripped model to 0.390, the same identity dependence found in §3, now on external data. **Injection transfer is a scale effect.** 2B fails outright (0.128 AUPRC, below the 0.150 base rate, AUROC 0.348); 8B reaches 0.797 AUPRC and 0.916 AUROC zero-shot; the rule baseline sits between the

Table 8: AUPRC on AgentDojo banking traces, zero-shot, $n=4,215$ ($n=4,133$ for injection). untrusted_destination has positives only and is excluded from macro. Guardian and agent-self rows are on a fixed 1,000-row subsample (positive rates 0.155 / 0.269 / 0.246 / 0.357 / 0.091). Source: docs/results/agentdojo/*-report.md.

Arm	injection	scope	policy	financial	destructive	macro
Rule baseline	0.250	0.253	0.237	0.896	1.000	0.527
Decoder 2B v4	0.128	0.257	0.849	1.000	1.000	0.647
2B v4, eval-stripped	0.125	0.218	0.694	1.000	1.000	0.608
2B v4, train-stripped	0.150	0.234	0.390	1.000	1.000	0.555
Decoder 8B v4	0.797	0.386	0.560	0.924	1.000	0.733
8B v4, eval-stripped	0.743	0.266	0.404	0.934	1.000	0.669
Granite Guardian 3.3 8B	0.103	0.289	0.224	0.257	0.217	0.218
Agent-self 8B (ALLOW/STOP)	0.291	0.442	0.449	0.591	0.105	0.375

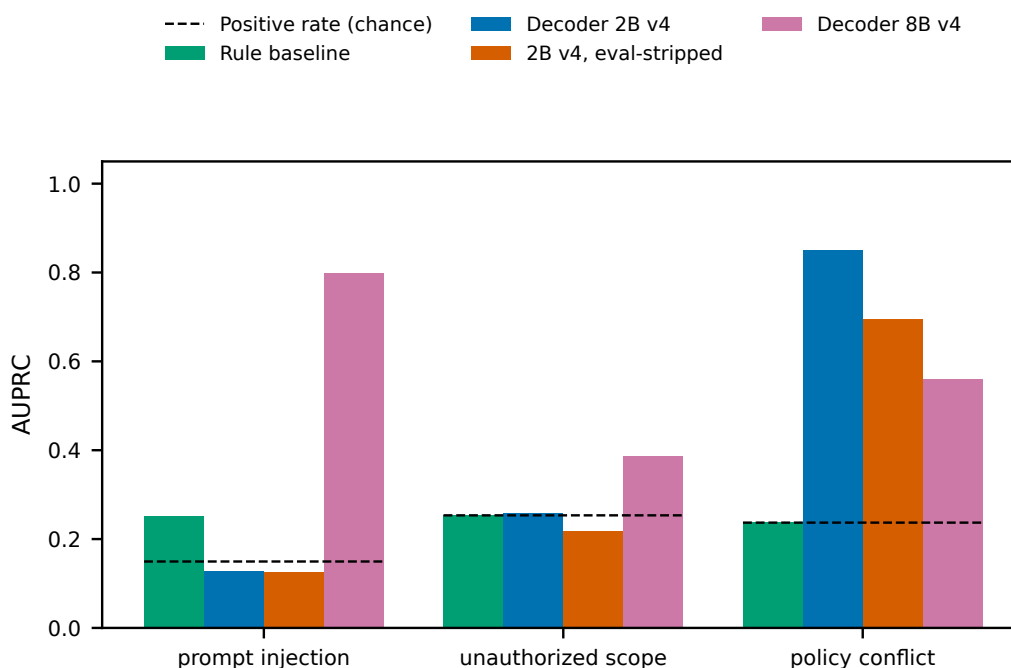


Figure 6: AUPRC on AgentDojo banking traces for the three learned dimensions, rule baseline vs. decoder 2B v4 vs. 2B v4 eval-stripped vs. decoder 8B v4, zero-shot, with each dimension’s positive rate marked by a dashed line. Source: docs/results/agentdojo/*-report.md.

two (0.250, AUROC 0.737) because presence of untrusted content alone is a strong but insufficient cue. This is the most important number in this section: the 8B checker, trained only on synthetic data, ranks the call that executes the attacker’s goal above the benign calls in the same poisoned trace 92 percent of the time. The 2B result is a plain failure, and a per-example score dump says why: the 2B injection head never fires on this domain. Raw logits for all 4,133 evaluable calls lie between -5.0 and -3.0 , the highest-scoring positive is -3.25 , and no call scores above zero; the residual ordering is noise. On the same rows the 2B `policy_conflict` head separates cleanly (median raw logit $+2.5$ for positives, -3.5 for negatives). The 2B model learned injection from surface cues of the synthetic generator that AgentDojo’s injections do not carry. The two 8B baselines separate scale from training: the same Granite-3.3-8B base asked ALLOW/STOP as the agent (§6) scores 0.291 AUPRC on injection and Granite Guardian 3.3 8B zero-shot scores 0.103, below base rate. The trained checker’s 0.797 is the training, not the parameter count. Stripping identity from the 8B input keeps injection (0.743) and drops `policy_conflict` from 0.560 to 0.404, the same content versus identity split as in §3. **unauthorized_scope does not transfer** (0.257 2B, 0.386 8B, vs. a 0.253 base rate); the label itself is the weakest one here, since "function not in the reference solution" counts harmless auxiliary reads as out of scope, and we report this without claiming transfer on it. **8B policy_conflict (0.560 AUPRC) is below 2B (0.849 AUPRC) while its AUROC (0.885) stays high**, which is unexplained; single-seed 8B vs. 2B comparisons on this cell fall within the seed noise §4 measured on synthetic heldout_policy_kind (sample sd 0.03) only for the AUROC gap, not for the AUPRC gap, so this remains a real open question. **Calibration does not transfer**: macro ECE runs 0.20–0.27 for the calibrated arms here, against under 0.03 on synthetic data, so the ranking transfers but the probabilities must be refit in-domain. Last, this is one suite (banking), one domain, with labels partly written by us (the four policy predicates) and partly by AgentDojo (prompt_injection_influence, unauthorized_scope); slack, travel, and workspace traces were not fetched.

9 RELATED WORK

LLM guardrails. Llama Guard (Inan et al., 2023), Granite Guardian (Padhi et al., 2024), Shield-Gemma (Zeng et al., 2024), NVIDIA Aegis (Ghosh et al., 2024), and gpt-oss-safeguard (OpenAI, 2025) are general-purpose content/harm classifiers over a conversation turn or action; Agent Action Guard (Pro-GenAI, 2025) is the same idea at small scale, a sentence-embedding MLP over the tool name and arguments alone (260 labelled rows, one per tool, no identity or policy input). We do not claim they fail at their intended purpose; §3 shows they are near chance specifically on delegated-authority dimensions (`unauthorized_scope`, `policy_conflict`, `insufficient_context`) they were not trained to answer, regardless of whether identity context is in their prompt.

Agent safety benchmarks. AgentHarm (Andriushchenko et al., 2024), ToolEmu (Ruan et al., 2023), InjecAgent (Zhan et al., 2024), and AgentDojo (Debenedetti et al., 2024) measure end-to-end agent behavior or injection success via an outcome judge, not principal entitlements or delegated scope as structured input, and produce no per-action calibrated probability a downstream policy engine can compose (§2, §5): they answer "did the agent do something bad," not "should this action have been allowed given who it was for."

Authorization and policy engines. OPA (Open Policy Agent), Cedar (Cedar Policy), and XACML (OASIS, 2013) are deterministic, symbolic engines: given a fully specified request and policy, they decide allow/deny with no learned uncertainty. forecheck’s policy layer is architecturally in this family, but the composition problem in §5 exists only because its input is a learned, calibrated probability rather than a fact a symbolic engine evaluates exactly.

Declared intent and deterministic enforcement. The OWASP agentic security guidance (OWASP GenAI Security Project, 2025) and practitioner positions built on it (CapiscIO, 2026) hold that an authorization decision must be deterministic and auditable, so a model’s judgment that an action “looks fine” cannot be the control; the agent should declare a typed mandate up front and a policy enforcement point should check each call against it symbolically. Our results support the first half: the acting agent is at chance on its own scope (§6). They also show the limit of the second: once the mandate is a natural-language policy, the symbolic checker (our rule baseline) drops to 0.33–0.50 AUPRC on `policy_conflict` and `unauthorized_scope`, and only a learned reader recovers it across paraphrases (§4). forecheck keeps the decision deterministic and puts the learned component where OWASP allows it: the scorer never authorises, it prices the review, and §5 measures what that advisory signal is worth.

Calibration. Platt/temperature scaling (Platt, 1999), the modern re-examination of network calibration (Guo et al., 2017), and the Wilson score interval (Wilson, 1927) (used for the approval-elimination bound, Appendix A) are standard tools we apply directly to policy *composition*, not where this literature is usually pointed: §5’s k -invariance result is a direct function of the underlying model’s calibration sharpness.

10 LIMITATIONS

Synthetic data only. Every number here is on data from forecheck’s own scenario generator (`docs/evaluation-plan.md`, classes 1–2). No human-labelled or production-traffic evaluation exists at submission time, and no real-world safety claim is made: every result measures whether an arm learned the generator’s own label function.

Independent labels exist for label validity, not for real-world risk. §7 adds a blind LLM judge as an independent reader of the same rendered text, which is a check on whether a label is observable in the text at all, not a check against real-world ground truth – the judge never saw production traffic either. That check is reassuring on `policy_conflict` ($\kappa=0.71$) and directionally consistent on `unauthorized_scope` ($\kappa=0.39$, differing definitions), but finds `insufficient_context` ($\kappa=0.06$) unrecoverable from the text by an independent reader. Every `insufficient_context` number in this paper should be read with that caveat attached until the generator’s rendering makes the context gap explicit.

Generator regularities. The model may be exploiting template regularities in how the generator writes policy clauses and entitlement lists, rather than a general policy-reading ability.

The `heldout_policy_kind/heldout_policy_phrasing` splits (§4) probe this and only partly rule it out: phrasing generalizes cleanly, kind does not, consistent with a lookup over kinds rather than the underlying operation. The encoder’s earlier null result on both splits is uninformative the same way – it never learned `policy_conflict` at all, so a flat curve shows nothing.

privilege_escalation **render** **defect** **(v2/v3).** The `authority_before/authority_after` fields deciding this label were never rendered into the tool-call text in v2/v3, so every arm’s 0.46–0.53 AUPRC there reflects a data defect: 78 of 79 dev positives were textually indistinguishable from an in-scope re-grant. Fixed in v4 (§4); numbers sourced from v2/v3 elsewhere should be read with this in mind.

Seed variance (three seeds each for v4 and v5). Three training seeds for decoder v4 agree on macro AUPRC to within 0.001 on every split (e.g. 0.960, 0.959, 0.960 on test). But the `heldout_policy_kind policy_conflict` cell moves across a 0.06 range between seeds (0.752, 0.689, 0.711; mean 0.717, sample sd 0.032) – an order of magnitude more than the macro noise floor – so differences smaller than that on this specific cell are not distinguishable from seed noise. Three seeds of the v5 (30-kind) recipe show the same range (0.704, 0.755, 0.717; mean 0.725, sample sd 0.026), which is what lets §4 compare the two bands directly instead of a single point each.

Held-out-policy-kind failure (now tested at 30 kinds and 8B). `policy_conflict` AUPRC drops from ~0.98–0.99 (in-distribution, unseen-phrasing) to 0.65–0.76 on entirely unseen kinds at 2B. The v5 experiment (ADR 0011, 30 kinds, 4 withheld) has landed across three seeds and is negative: mean 0.725, inside one sample sd of v4’s mean 0.717 – tripling the kind inventory did not fix this at 2B. An 8B LoRA on the unchanged v4 data reaches 0.851, so capacity helps partially but does not solve it. The supportable claim today is "one model, policies of a kind seen in training (or with enough capacity to partially generalize), any phrasing" – not "arbitrary new policy kinds."

External result is one suite, partly self-written labels. The AgentDojo transfer result (§8) covers a single suite (banking), a single domain, and labels that are partly written by us (the four policy predicates) and partly by AgentDojo (`prompt_injection_influence`, `unauthorized_scope`); slack, travel, and workspace traces exist in AgentDojo but were not fetched. Calibration did not transfer at all: macro ECE runs 0.20–0.27 on AgentDojo against under 0.03 on synthetic data, so only the ranking, not the probabilities, generalizes to this external distribution.

Other caveats. A single generator family (English, template renderer) produces every example. The §5 cost table is the default severity tier, not tuned to any organization. Generator seed nondeterminism was found and partially fixed mid-project (`docs/results/notes/privilege-escalation-diagnosis.md`); v2/v3 data are not byte-reproducible from their configs.

11 CONCLUSION

Pre-execution risk for agent tool calls is mostly a delegated-authority problem, not a content-safety problem, and general-purpose safety models do not solve it: stripping identity and policy context from a trained classifier’s input collapses exactly the dimensions that depend on it, and two zero-shot safety models are at chance on those same dimensions whether or not the context is available to them. Identity/delegation context and calibration quality are not incidental engineering details for agent guardrails – they are the two properties that determine whether a guardrail can detect authorization violations at all (§3) and survive being deployed alongside more than one policy (§5). What remains open is whether policy-conditioned reasoning generalizes to genuinely novel policy kinds, not just novel wording of familiar ones (§4); the v5 experiment testing this, and a first real-data probe against hand-labelled traffic, are the next things this line of work needs before any claim above can be extended past synthetic data. The AgentDojo transfer result (§8) is a first step on that last point: the policy-conflict and prompt-injection rankings hold zero-shot on agent traces and policies forecheck never saw, but the probabilities do not, so calibration remains a synthetic-data property until it is refit in-domain.

REFERENCES

- Maksym Andriushchenko, Alexandra Souly, Mateusz Dziemian, Derek Duenas, Maxwell Lin, Justin Wang, Dan Hendrycks, Andy Zou, Zico Kolter, Matt Fredrikson, Eric Winsor, Eliot Wynne, Yarin Gal, and Xander Davies. Agentharm: A benchmark for measuring harmfulness of llm agents. *arXiv preprint arXiv:2410.09024*, 2024.
- CapiscIO. Automated intent classification for AI agents is an unsolved problem. <https://capisc.io/blog/automated-intent-classification-for-ai-agents-is-an-unsolved-problem>, 2026. Vendor position piece. Accessed 2026-09-22.
- Cedar Policy. Cedar policy language documentation. <https://www.cedarpolicy.com/en/documentation>. Accessed 2026-09-22.
- Edoardo DeBenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. Agentdojo: A dynamic environment to evaluate prompt injection attacks and defenses for llm agents. *arXiv preprint arXiv:2406.13352*, 2024.
- Shaona Ghosh, Prasoon Varshney, Erick Galinkin, and Christopher Parisien. Aegis: Online adaptive ai content safety moderation with ensemble of llm experts. *arXiv preprint arXiv:2404.05993*, 2024.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- IBM Granite Team. Granite 3.0 / 3.3 language models: Model card and technical report. Technical report, IBM Research, 2024. See also Granite 3.3 (2025) release notes.
- Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, and Madian Khabza. Llama guard: Llm-based input-output safeguard for human-ai conversations. *arXiv preprint arXiv:2312.06674*, 2023.
- OASIS. extensible access control markup language (xacml) version 3.0. OASIS Standard, <http://docs.oasis-open.org/xacml/3.0/xacml-3.0-core-spec-os-en.html>, 2013.
- Open Policy Agent. Open policy agent (opa) documentation. <https://www.openpolicyagent.org/docs/latest/>. Accessed 2026-09-22.
- OpenAI. gpt-oss-safeguard: Open-weight reasoning models for safety policy classification. Technical report, OpenAI, 2025. Model and technical report.
- OWASP GenAI Security Project. OWASP securing agentic applications guide. <https://genai.owasp.org/>, 2025. Accessed 2026-09-22.
- Inkit Padhi, Manish Nagireddy, Giandomenico Cornacchia, Subhajit Chaudhury, Tejaswini Pedapati, Pierre Dognin, Keerthiram Murugesan, Erik Miehling, Martin Santillan Qian, Ramya Ramamurthy, Giulio Zizzo, Ambrish Eltoukhy, Michael Hind, Prasanna Ram, Payel Das, Pin-Yu Chen, Jerret Cunningham, Ananya Agarwal, Prasanna Sattigeri, Kush Raghavan, and Kush R. Varshney. Granite guardian. *arXiv preprint arXiv:2412.07724*, 2024.
- John Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*, pp. 61–74, 1999.
- Pro-GenAI. Agent action guard: Classifying AI agent actions to ensure safety and reliability. Preprint, <https://doi.org/10.20944/preprints202510.1415.v2>; code <https://github.com/Pro-GenAI/Agent-Action-Guard>, 2025. Accessed 2026-09-22.

Table 9: Multi-policy stacking, `heldout_family` ($n=6640$). Source: `docs/results/synthetic-v2/README.md, docs/evaluation/multi-policy-stacking.md`.

Arm	k	Strategy	FPR	FNR	Review rate
Decoder 2B	1	any	0.015	0.253	0.31
Decoder 2B	2	independent	0.140	0.080	0.30
Decoder 2B	2	expected_cost_joint	0.082	0.070	0.28
Decoder 2B	3	independent	0.966	0.016	0.66
Decoder 2B	3	joint	0.905	0.051	0.63
Decoder 2B	3	expected_cost_joint	0.082	0.070	0.28
Decoder 2B	8	independent	0.966	0.007	0.58
Decoder 2B	8	expected_cost_joint	0.082	0.070	0.28
Encoder granite-r2	2	independent	0.710	0.042	0.26
Encoder granite-r2	2	expected_cost_joint	0.921	0.010	0.37
Encoder granite-r2	3	independent	0.990	0.008	0.21
Encoder granite-r2	3	expected_cost_joint	0.921	0.010	0.37

Yangjun Ruan, Honghua Dong, Andrew Wang, Silviu Pitis, Yongchao Zhou, Jimmy Ba, Yann Dubois, Chris J. Maddison, and Tatsunori Hashimoto. Identifying the risks of lm agents with an lm-emulated sandbox. *arXiv preprint arXiv:2309.15817*, 2023.

Benjamin Warner, Antoine Chaffin, Benoît Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Nathan Cooper, Griffin Adams, Jeremy Howard, and Iacopo Poli. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. *arXiv preprint arXiv:2412.13663*, 2024.

Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

Wenjun Zeng, Yuchen Liu, Ryan Mullins, Ludovic Peran, Joe Fernandez, Hamza Harkous, Karthik Narasimhan, Drew Proud, Piyush Kumar, Bhaktipriya Radharapu, Olivia Sturman, and Oscar Wahltinez. Shieldgemma: Generative ai content moderation based on gemma. *arXiv preprint arXiv:2407.21772*, 2024.

Qiusi Zhan, Zhixiang Liang, Zifan Ying, and Daniel Kang. Injecagent: Benchmarking indirect prompt injections in tool-integrated large language model agents. *arXiv preprint arXiv:2403.02691*, 2024.

A APPENDIX

A.1 FULL MULTI-POLICY STACKING TABLE

Table 9 extends Table 4 with the `joint` strategy (rules from all k policies merged into one bundle and evaluated once in threshold mode, rather than decided independently and OR-ed), false negative rate, review rate, and the `granite-embedding-r2` encoder run on the identical examples and synthetic per-dimension policies. `independent` and `joint` are mathematically identical at every k here because the synthetic stacking policies use only plain, non-hard, non-allow_override rules, so there is nothing for the engine’s combination logic to do differently between the two strategies on this construction (`docs/evaluation/multi-policy-stacking.md` §3); the two rows diverge only once real bundles with `hard/allow_override` rules are mixed in, which is outside this diagnostic.

Note the encoder’s `expected_cost_joint` FPR (0.921) is *worse* than its own `independent` FPR at $k=2$ (0.710): on a poorly calibrated probability vector, aggregating across dimensions does not reliably help and can hurt, underscoring that the composition benefit in §5 is conditional on calibration quality, not a property of the `expected_cost` rule in isolation.

Table 10: Decoder 2B, approval elimination on heldout_family (v2 $n=6640$, v4 $n=6653$), bundle balanced. “raw” is the split’s own 57–58% base incident rate; “rw” (reweighted) importance-reweights rows so the weighted incident rate equals 5%, matching a plausible deployment rate. Source: docs/results/synthetic-v2/README.md, docs/evaluation/approval-elimination.md.

Budget	Elim. (v2, raw)	Elim. (v2, rw)	Elim. (v4, rw)	Review (v4, rw)
0.1%	0.000	0.000	0.000	0.969
0.5%	0.000	0.000	0.000	0.969
1%	0.000	0.888	0.903	0.067
2%	0.000	0.961	0.952	0.021
5%	0.000	0.992	0.992	0.001

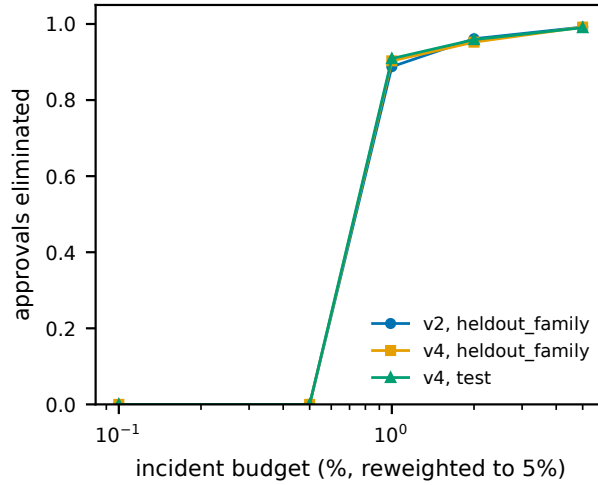


Figure 7: Approvals eliminated vs. incident budget, reweighted to a 5% deployment incident rate, for decoder 2B v2 (heldout_family, $n=6640$) and v4 (heldout_family $n=6653$, test $n=4412$), all synthetic data, log-scaled budget axis. Source: docs/results/synthetic-v2/decoder-2b/heldout_family-approval-report.md, decoder-2b-v4/{heldout_family,test}-approval-report.md.

A.2 APPROVAL ELIMINATION

The product-facing metric is not classifier accuracy but how many of a deployment’s human-approval reviews can be safely automated at a fixed, chosen incident rate: sort examples by the decoder’s expected cost of ALLOW, and ask what fraction of the review queue can be auto-decided while the fraction of auto-allowed actions that are actually risky stays under a budget. Because the evaluated heldout_family split is adversarial-heavy (58% base incident rate), Table 10 reports both the raw unweighted curve and a curve importance-reweighted to a stated 5% deployment incident rate, with elimination at each budget computed conservatively via a 95% Wilson score upper bound (Wilson, 1927) on the realized incident rate in the auto-allowed set.

Figure 7 shows the three curves nearly overlapping: v2 and v4 eliminate a comparable share of approvals at every budget on heldout_family, and v4’s test split tracks its own heldout_family curve closely, so the elimination curve is not sensitive to the v3/v4 generator fix or to family-held-out tools.

At a deployment incident rate of 5%, the calibrated v2 decoder auto-allows 88.8–99.2% of actions that would otherwise need human review while keeping the allowed-but-risky rate at or below the stated budget. The unweighted column is near zero not because the model is uninformative but because the split’s own top-ranked rows already contain risky examples by construction (an adversarial benchmark, not a deployment population); it is a ceiling of the split’s design, not of the classifier.

972 Below a 1% budget the Wilson bound cannot be met at this split's effective sample size and the
973 conservative curve reports zero elimination; a point-estimate mode is available for that regime but is
974 dominated by single rows at small allow-set sizes.

975
976 A.3 EXPERIMENT MATRIX
977

978 The full list of experiments this paper still needs, with recipe files and status, is tracked in
979 `paper/EXPERIMENTS.md` rather than duplicated here.

980
981
982
983
984
985
986
987
988
989
990
991
992
993
994
995
996
997
998
999
1000
1001
1002
1003
1004
1005
1006
1007
1008
1009
1010
1011
1012
1013
1014
1015
1016
1017
1018
1019
1020
1021
1022
1023
1024
1025